

Lerato Emmanuel Mgwangqa

Junior Software Engineer and Data Analyst | Specializing in Full Stack Development, AI/Machine Learning, and Networking

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PROFESSIONAL SUMMARY

A driven and skilled BSc graduate specializing in Mathematical and Computer Sciences, with hands-on experience in software development, data analysis, and networking. Proficient in Python, C/C++, and R, with a strong foundation in full-stack development, backend technologies, and machine learning. Proven track record as a peer mentor, where I enhanced student performance through organized study support and personal mentorship. Recognized for problem-solving abilities, attention to detail, and dedication to continuous learning. Prepared to contribute effectively in roles that require technical acumen, analytical thinking, and project execution in software engineering and data-driven environments.

EDUCATION

BSc in Mathematical and Computer Sciences (January 2022 - November 2024)

Sol Plaatje University, Kimberley

Graduation: **25 APRIL 2025**

Completed coursework encompassing a broad spectrum of topics in computer science, mathematics, and statistics, including:

Computer Organization, Computer Architecture, Programming, Data Structures and Algorithms, Algorithm Analysis and Program Design, Operating Systems, Artificial Intelligence, Database Systems. Mathematical foundations in Algebra, Linear Algebra, Calculus, Advanced Calculus, Probability Theory, and Mathematical Analysis. Statistical expertise in Statistical Inference, Regression Analysis, Non-Parametric Statistics, Time Series Analysis, and Generalized Linear Models. Proficiency in interdisciplinary topics such as Numerical Methods, Mathematical Modelling, and Markov Processes.

ALX Software Engineering Program (January 2023 - July 2024)

African Leadership Experience (ALX), Online

Completed a comprehensive program focused on modern backend development techniques and tools.

Key areas of expertise include:

Languages & Frameworks: Modern JavaScript (ES6+), TypeScript, Node.js, Express.js, Python, C

Backend Development: Advanced MySQL, NoSQL, Redis, Queuing systems, caching, pagination, user authentication, and data security

DevOps & Systems Engineering: Linux basics, web servers (Nginx), application servers, load balancing, monitoring, CI/CD, and web stack debugging

Matriculation

Dinokaneng Secondary School, 2021

EXPERIENCE

January 2024 – November 2024

Peer Mentor, Sol Plaatje University

- Provide academic and personal support to over 20 students, enhancing their academic performance and study skills.
 - Organize study groups and workshops, contributing to a collaborative learning environment.
 - Collaborate with faculty to develop action plans for struggling students, supporting a retention increase within the program.
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SKILLS

Programming Languages: Python, C/C++, R, Shell Scripting

Web Development: HTML5, CSS3, JavaScript (Express.js, Node.js), React.js

Databases: MySQL, MongoDB, Redis

Frameworks & Tools: Flask, Streamlit, Scikit-Learn, NLTK, Jupyter Notebook, Joblib, Unit-Testing

Version-Control: Git, GitHub, Linux (ubuntu wsl), vi&vim

Networking Skills: Ethernet, Wi-Fi, VLAN, NAT, TCP/IP, DNS, DHCP, HTTP/HTTPS, FTP/SFTP, SSH

Soft Skills: Technical Writing, Collaboration, Time Management, Public Speaking, Analytical Thinking, Resilience, Excellent computer skills (Microsoft skills- Excel, Powerpoint, Word.....etc)

PROJECTS

GitHub Top Repo Analysis

- Description: Analyzed GitHub's top repositories to identify trends in programming languages and technologies.
- Tools Used: Python, Streamlit, GitHub API, Data Analysis Techniques

Residence-Based Management System

- Description: Developed a system to streamline and manage student residence data, improving administrative efficiency.
- Tools Used: MySQL, HTML5, CSS3, JavaScript

C Printf Implementation

- Description: Created a simplified version of the C printf function to deepen understanding of C's low-level functionality.
- Tools Used: C Programming Language

Chatbot Project

- Description: Built a chatbot for intent classification and NLP, using machine learning for enhanced user interaction.
- Tools Used: Python, Flask, Scikit-Learn, NLTK, Pandas, HTML, CSS

E-Learning Platform for Skill Development

Learnaz Hub

- Description: An engaging e-learning platform offering courses across various fields, designed to make learning accessible for all ages.
 - Tools Used: React, TailwindCSS, Node.js, Express, MongoDB, Git, GitHub,
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CERTIFICATIONS

Full Stack Software Engineering – *ALX* Software Engineering Program (Graduation: August 2024)
[Certificate Awarded](#)

Peer Mentor, *Sol Plaatje University*- Certification of Completion (honored on 30 Nov 2024)

[Peer Mentor Certificate 2024 - Lerato Mgwangqa](#)

REFERENCES

Available upon request.
